



Guangdong Technion

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Homework 9 - Deadline January 3rd

1. Let $a_n = n^{\frac{1}{n}} : n \geq 2$ and $f(x) = \frac{\sin(x^2 - 1)}{(x - 1)}$ for $x \neq 1$. Find $\lim_{n \rightarrow \infty} f(a_n)$.

2. Decide if each of the following statements are true or false. In the true case, justify; in the false case, give a counterexample.

a) If $\lim_{n \rightarrow \infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ is convergent.

b) $\sum_{n=1}^{\infty} \frac{2^n + 4^n}{\pi^n}$ is divergent.

c) $\sum_{n=1}^{\infty} \sqrt[3]{n \sqrt{\frac{2}{n^4}}}$ is divergent.

3. Analyze the convergence of the series

a) $\sum_{n=2}^{\infty} \frac{1}{n \ln(n)},$

b) $\sum_{n=2}^{\infty} \frac{1}{n [\ln(n)]^n}.$

4. a) Approximate the series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ by using the sum of the first 10 terms. Estimate the error involved in this approximation.

b) How many terms are required to ensure that the partial sum S_N of the series

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$$

is exact to within 5×10^{-4} ?